1 A student is investigating the force between two charged metal spheres S and T, as shown in Fig. 1.1.

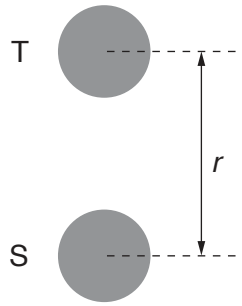


Fig. 1.1

Each sphere may be charged by connecting the positive lead from a power supply to the sphere and then removing the lead. The electromotive force (e.m.f.) of the power supply used to charge sphere T is V.

The force F between the two charged spheres may be determined by attaching sphere S to a top pan balance.

For a constant charge on sphere S, it is suggested that the relationship between F and V is

$$F = \frac{\alpha V}{r^2}$$

where r is the distance between the centres of the spheres and α is a constant.

Design a laboratory experiment to test the relationship between F and V.
Explain how your results could be used to determine a value for α.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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